

Harsh Vardhan Tripathi

+91 95652 49247 | harsh.tripathi.cs@gmail.com | Lucknow, India | linkedin.com/in/iamharshvardhantripathi | github.com/itripathiharsh

Summary

AI/ML Engineer specializing in **LLM application development**, **retrieval-augmented generation (RAG) pipelines**, and **agentic AI systems**. Skilled in fine-tuning transformer models, optimizing inference costs, and deploying production-ready AI systems using Python, PyTorch, LangChain, and cloud APIs. Currently building scalable LLM-powered features in a live professional product environment.

Experience

AI Developer Intern

Banao Technology

Nov 2025 – Present

Lucknow, India

- Reduced LLM inference costs by **20%+** through prompt engineering, response caching, and intelligent provider routing (Groq, Gemini, OpenAI) based on task complexity and latency requirements.
- Designed and evaluated **retrieval-augmented generation (RAG)** pipeline components for client-facing products, improving answer relevance and measurably reducing model hallucinations.
- Shipped **3+ agentic AI features** to production collaborating with cross-functional AI and backend teams; contributed to scalable architecture via model performance benchmarking and token usage analysis.

AI Intern

Innovate

Jun 2025 – Jul 2025

India (Remote)

- Built a **fully offline local chatbot** (Qwen 0.5B LLM) with 4-turn sliding memory window, achieving **2–5s CPU inference latency** with zero GPU dependency using Hugging Face Transformers and PyTorch.
- Deployed dual-interface (CLI + Streamlit web UI) with modular architecture; optimized model loading and memory management to reduce cold-start time in a resource-constrained environment.
- Documented the system architecture and deployment process, enabling straightforward onboarding and extension for future team members.

Machine Learning Intern

Edunet Foundation (AICTE Partnered)

Apr 2025 – May 2025

India (Remote)

- Trained **ResNet50** via transfer learning for automated plant disease detection, achieving **95% accuracy across 38 classes** on a curated and cleaned dataset of **50,000+ images**.
- Built end-to-end deep learning data pipeline with augmentation, normalization, and class-balanced batching; documented precision, recall, and F1-score metrics for evaluation and stakeholder reporting.
- Presented findings and model performance analysis to program supervisors, demonstrating the pipeline's scalability for real-world agricultural disease monitoring applications.

Projects

Green Minds – AI Wellness Journal | Streamlit, Firebase, Hugging Face, Groq, Gemini

2024

- Fine-tuned **BERTweet + GoEmotions** for multi-label emotion recognition, achieving **92% F1-score** — a 7% improvement over baseline — deployed within a production mental wellness platform.
- Integrated **Groq (Llama 3) + Gemini API** via prompt-engineered agentic workflows to deliver real-time reflective guidance and 30+ interactive wellness activities to active end users.
- Architected full-stack platform on Streamlit + Firebase with daily mood analysis, persistent journaling, and real-time AI feedback; actively maintained in production with real user traffic.

EcoRAG Agent – Environmental RAG System | LangGraph, ChromaDB, Groq Llama 3

2024

- Engineered a fault-tolerant **climate-focused RAG system** using Groq Llama 3.1 8B with automatic multi-key failover, sustaining **99.9% uptime** on a live Hugging Face Spaces deployment.
- Built a 4-step **LangGraph agentic pipeline** (Retrieve → Refine → Answer → Validate) with MiniLM-L6-v2 sentence embeddings and ChromaDB as the vector store backend.
- Eliminated gRPC module conflicts to achieve **cold start under 15s** and **2–4s CPU inference**, enabling fully viable deployment on CPU-only free-tier infrastructure.

Technical Skills

Languages: Python, SQL (PostgreSQL), C

ML / AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph, Fine-tuning, Transfer Learning

LLM & Agentic AI: Prompt Engineering, RAG Pipelines, Agentic Workflows, Groq API, Gemini API, OpenAI API, Llama 3

NLP / Computer Vision: BERT, BERTweet, GoEmotions, OpenCV, MediaPipe, ResNet, TF-IDF

Vector Stores & Databases: ChromaDB, FAISS, PostgreSQL, SQLite

Backend / Cloud / MLOps: FastAPI, Streamlit, Firebase, Google Cloud, Docker, GitHub Actions, CI/CD, Render

Education

Babu Banarasi Das Institute of Technology & Management (BBDITM) <i>B.Tech in Computer Science and Engineering</i>	Lucknow, India <i>Aug 2022 – May 2026 (Expected)</i>
Central Board of Secondary Education (CBSE) <i>Senior Secondary (Class XII)</i>	Lucknow, India <i>Mar 2021</i>